F876 — Lane A, my half (blind rephasing-invariance check + honest tier): the checker duty PASSES (J is a genuine physical CP invariant, rephasing-dev 1e-17), the CP MAGNITUDE is ~650× observed (existence-only, corroborates Lane B), and — the honest catch — my BLIND prediction that the angle ordering would be forced is REFUTED by my own toy: a crude rank-2 FK overlap gives O(45–69°) angles, NOT the small hierarchical CKM. I do NOT claim the angles come out; that is Elie's real-localization computation. Calibrate both ways: I did not over-claim the magnitude (predicted off, confirmed), and I DID over-predict the angle structure — the number corrected me.

Lyra, Sunday 2026-08-09. Lane A checker duty. Pre-registered blind (F876_laneA_blind_prereg.md) BEFORE any numpy, then computed. RUBRICS done-bar below.

What I committed blind, and what the number said

blind prediction	result	verdict
P1 Jarlskog J rephasing-invariant under D_L V D_R	max|dJ| = 1.4e-17 over 8 rephasings	PASS ⍰ — J is physical, not gauge
P2 closed-form overlap = brute Bergman integral	MC/CF ratio ~0.60 but phase off 0.80 rad (crude 1-cdim slice)	INCONCLUSIVE — my MC measure too crude; hand to Elie's integrator
P3 angle ordering $\theta_{12}>\theta_{23}>\theta_{13}$ forced	angles = 65.1° / 43.3° / 69.0° (unordered, huge)	REFUTED × — my over-prediction; toy does NOT force CKM
P4 J magnitude ~O(100×) off (existence only)	J = −2.0e-2 vs observed 3.08e-5 → 650×	confirmed direction ⍰ (worse than 100×)

The two things that BANK from my side (and only these two)

1. Rephasing-invariance holds (blind-committed, PASS). J = $\mathrm{Im}(V_{01}V_{12}V_{02}V_{11})$ is invariant under diagonal field rephasings to 1e-17. So the CP violation the geometry produces is a genuine physical invariant, not a removable phase convention. This is the load-bearing checker result and it is clean.
2. The CP MAGNITUDE is OFF (~O(100–650×)), existence only. My toy gives |J| ~ 2e-2 vs observed 3.08e-5. This corroborates Lane B's ~O(100×) statement and Grace/Cal's J≈2.6e-3. The CP-phase magnitude is NOT forced-correct; it stays an open item. Every δ_CP / J value is a reverse-fit (K684).

What does NOT bank (my over-prediction, owned)

- The mixing ANGLES do not come out of my toy. I predicted (blind) the ordering would be forced; the crude FK overlap gave O(45–69°) angles, nowhere near the small hierarchical CKM (13.0°/2.4°/0.2°). The reason is physical and it is the real content of #68: the smallness of the CKM angles requires the up and down radial towers to be NEARLY ALIGNED (near-diagonal overlap), and my stand-in radii/direction (arbitrary u_0, scale 0.55) do not encode that alignment. Whether the ACTUAL derived Yukawa localizations produce near-alignment (small angles) is exactly what Elie's real overlap must show. My toy neither confirms nor refutes that — it only refutes my premature "ordering forced" claim.
- The brute-integral agreement is not yet shown — my 1-cdim MC slice with a guessed Bergman weight is too crude (phase off). The reproducing-property check needs the proper D_IV^5 measure; that is Elie's integrator, not my scaffold.

The honest Lane-A verdict (my half)

The forced win is rephasing-invariance (J is physical) — that is real and blind-verified. The magnitude is off (~O(100×)), so CP stays existence-only, magnitude open — no change to the standing order. The angles are NOT forced by a generic FK overlap; their smallness is a specific near-alignment question that only the real derived-localization computation (Elie) can settle. I explicitly retract the pre-registered "angle ordering forced" expectation — the number killed it. The claim "the rank-2 overlap gives the CKM angles and the phase value in a single shot" is, from my computation, not yet demonstrated: the phase magnitude is wrong by ~650×, and the angles need the real localizations. This is the calibrate-both-ways result: the checker duty passed, the optimistic framing did not.

RUBRICS Layer-2 done-bar (checker duty, honest)

- ☒ Pre-registered all four predictions BLIND to disk before computing.
- ☒ Ran the number; P1 (rephasing-invariance) PASS at 1e-17 — the load-bearing check.
- ☒ P4 magnitude off confirmed (~650×) → existence-only holds; magnitude open.
- ☒ P3 REFUTED — owned my over-prediction; do NOT claim angles forced; located the real content (tower near-alignment).
- ☒ P2 inconclusive — flagged my MC as too crude; handed the proper-measure check to Elie.
- ☒ Tier: rephasing-invariance banked (J physical); magnitude OFF; angles NOT banked; nothing over-claimed.

Handoffs

- @Elie (Lane A partner — the real computation) — my half: (1) rephasing-invariance PASSES blind (J invariant to 1e-17) — J is physical, use it. (2) In my crude toy the CP magnitude is 650× observed and the angles come out O(45–69°) — so a generic FK overlap does NOT force the small CKM. The real question #68 must answer: do the ACTUAL derived Yukawa localizations make U_up and U_down nearly aligned (→ small hierarchical angles)? My stand-in radii don't, by construction. (3) Fire your overlap against the brute Bergman integral with the PROPER measure — my 1-cdim MC was too crude to confirm the reproducing property (phase off 0.8 rad). When you have V, I'll re-check rephasing-invariance and commit my half blind. Keep the magnitude OFF — J value is a reverse-fit; report angle STRUCTURE, not the phase value.
- @Cal (Lane B magnitude gate) — independent corroboration for your magnitude-honesty gate: my blind toy gives |J| ~ 2e-2 = ~650× observed (your ~O(100×) is if anything generous). The paper must state CP existence only, magnitude open, cite NO J value. I flagged this in the v0.2 draft (Section II.3, "magnitude OFF"); it's consistent with your gate.
- @Keeper — Lane A honest status from my side: rephasing-invariance banked (J physical); CP magnitude ~650× off → existence-only holds (no change to the ledger); the mixing ANGLES are NOT yet forced (my toy refuted my own optimism; the real test is Elie's localizations). The flagship's CP row (Structural, forced, magnitude OFF) stands exactly as written — this computation supports it, adds nothing to promote.
- @Casey — the finish-line computation, run honestly, and it came out half-and-half — which is the true state, so I'll give it to you straight. The good half: I checked, blind, that the CP violation our geometry makes is *real physics* and not a bookkeeping artifact — you can spin every field's phase freely and the asymmetry doesn't budge (invariant to 17 decimal places). That half is solid and banked. The honest half: I had predicted, before computing, that the same overlap would also line up the mixing *angles* in the right pecking order — and my own toy said no. It gave big, unordered angles (45–70°) instead of the tiny real ones (13°, 2°, 0.2°), and the CP size came out ~650× too large. So two things stay off the table: the CP *magnitude* (still a reverse-fit, ~100× problem, exactly what Cal's gate says) and the angle *values*. The real question I can now name precisely: the CKM angles are small because the up-quark tower and the down-quark tower sit *almost on top of each other* radially, and my crude stand-in positions don't capture that near-alignment — whether the true, derived quark positions do is Elie's computation to settle, not mine to assume. I over-predicted the angles; the number corrected me; I'm reporting the correction rather than the hope. Nothing pushed.

Notes only; no toy/theorem claimed (checker duty + honest tier). F876: Lane A my half. BLIND prereg (before compute). P1 rephasing-invariance: J invariant to 1.4e-17 → PASS, J physical. P4 magnitude: |J|=2e-2 vs 3.08e-5 = 650× → existence only, magnitude OFF (corroborates Lane B ~O(100×), Grace/Cal J~2.6e-3). P3 angle ordering forced: REFUTED — crude FK overlap gives 65°/43°/69°, not small hierarchical CKM; I retract my over-prediction; the smallness = tower near-alignment question = Elie's real-localization computation. P2 brute-integral: inconclusive (my 1-cdim MC too crude, phase off 0.8 rad) → Elie's proper-measure integrator. BANKS: rephasing-invariance (J physical) + magnitude-off (existence only). Does NOT bank: angles. Calibrated both ways; over-prediction owned. Nothing pushed. — Lyra